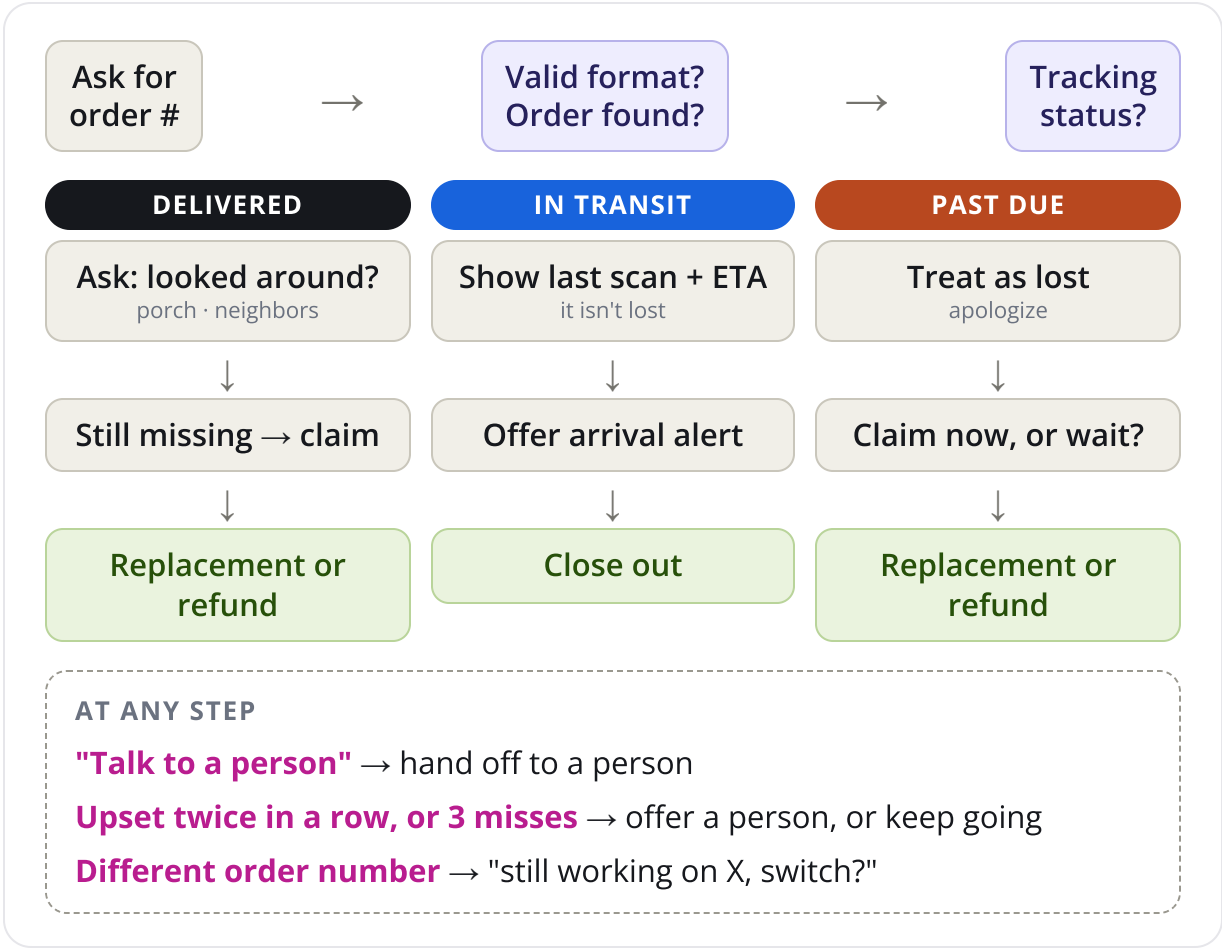


Work out what's really wrong first, then act

- 1 Triage before anything else.** "Lost package" is usually one of three situations, and only one is actually lost: *delivered but missing, still in transit, or past due with no movement*. Each gets a different path.
- 2 A designed flow with natural language on top.** Customers type however they talk. The steps are fixed, so every path is predictable and testable.
- 3 Errors handled by kind.** A bad format and an unknown order get different messages. Three unmatched messages, or two upset ones, offer a person. A new order number mid-chat asks before switching.
- 4 Never leave the customer hanging.** Every reply ends by asking what they want and naming the options.



Gemini decides and words it. The app holds the facts.

Browser · HTML / CSS / vanilla JS chat UI

Sends only the customer's text and a session id

↓ text only

Node + Express server · holds conversation state and the API key

Guardrails, rate limits, CORS, fallbacks

↓ each turn

engine.js · the flow

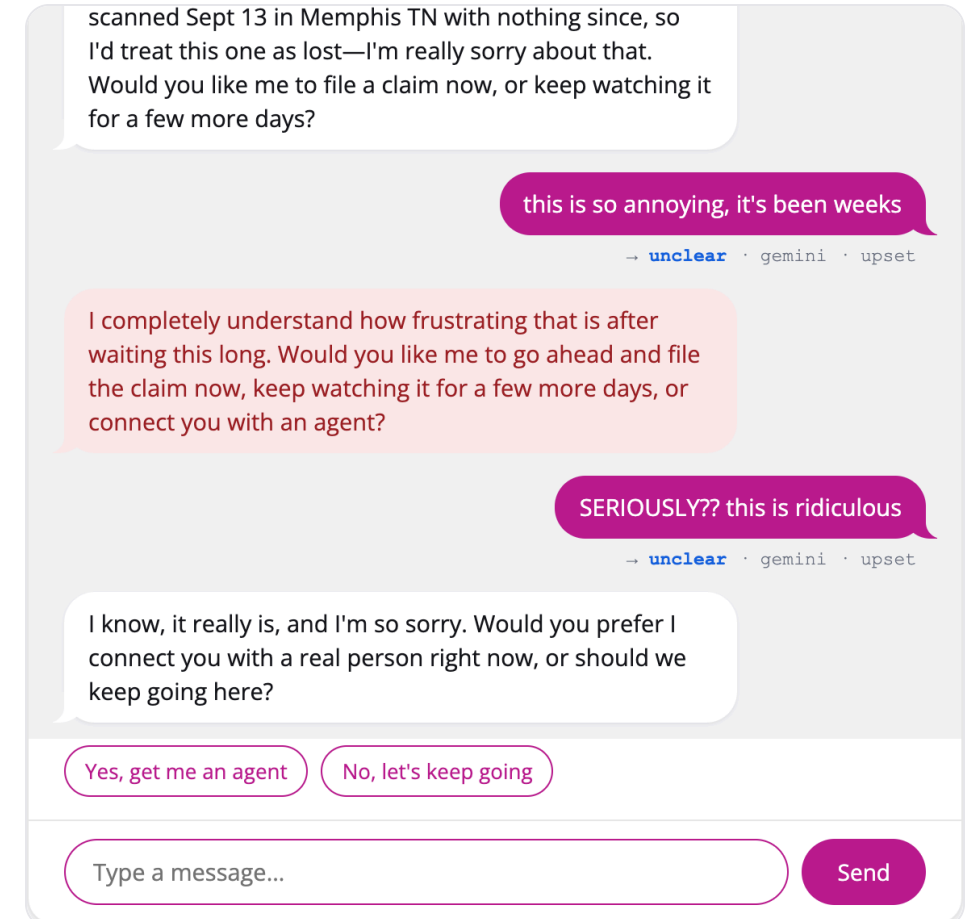
10 states, 19 moves, the facts, the draft wording

Gemini (function calling)

① pick one legal move ② reword the draft

- **Only legal moves are offered** to Gemini each step, and it must call exactly one. Illegal picks are rejected.
- **The reword is checked.** If it changes a number, date or order id, or drops the follow-up question, the plain draft is sent.
- **Graceful fallbacks.** Gemini down → keyword rules. Server down → the same engine runs in the browser.
- **120 automated tests** cover every order, error path, edge case and guardrail. Deployed free on Render.
- **orders.json / users.json emulate a database and an auth system** — no real infrastructure to stand up for the demo.

Node 18+ · Express · Gemini API · Render



Agent mode: each customer message shows the move chosen, who chose it, and an upset flag. Two upset in a row offers a person.

orders.json / users.json emulate a carrier database and an auth system

Making an AI natural without making it unsafe

Letting an LLM talk without letting it promise things

Split the jobs. The app owns state and facts; Gemini only picks a legal move and rewords a draft. Rewrites that change a fact are thrown away, and a manipulated model can only pick the wrong move from a short list.

Sounding human, but never leaving the customer stuck

Tuned by testing real chats. Early replies ended on statements, and frustration alone handed off to a person. Now every reply asks what to do next, and a person is offered only when asked, after 3 misses, or after 2 upset messages.

Messy real-world inputs

31 mock orders covering 11 statuses exposed gaps: a customer pasting a new order number mid-chat was "unclear", and a wrong-address delivery was told to check the door. Now the bot confirms before switching, and misdelivered orders go straight to a claim.

Hosting and connecting a free server

CORS, cold starts and file:// pages. A page opened from disk has a null origin, and the free Render tier sleeps for 30 to 60 seconds. The page falls back to keyword mode so the demo never breaks, and the debug panel explains why it fell back.

From prototype to something customers can rely on

Make it real

- **A real database and real auth**, replacing orders.json / users.json — hashed, salted passwords checked only server-side, not shipped to the browser the way offline mode's demo accounts are today.
- **Real carrier lookups** (USPS, UPS, FedEx) in place of the mock orders.
- **A true handoff**. Open a ticket or live chat and pass the full transcript so customers never repeat themselves.
- **Claims and alerts that actually happen**: file the claim, send the email or SMS.

Make it safer and smarter

- **1-2 backup AI providers** tried between Gemini and keyword rules — a second model or provider before falling all the way back — for uptime through a single provider's outage, with keyword matching as the true last resort.
- **Check meaning, not just numbers**, when validating Gemini's rewording.
- **An evaluation set** of messy real phrasings to score moves before each change.
- **Persistent sessions**, so a restart or a sleeping server doesn't lose the chat.

WHAT I'D MEASURE

Where customers drop out of the flow, how often the bot resolves without a person, how often it falls back to keyword rules, and how often Gemini's rewording is rejected.